



DPUVDSSCNX

This DPU collects the height of ground at up to six (wheel) positions from the SCNX database. Positions are given by coordinates of the SILAB world coordinate system.

1. Inputs:

<i>Xfl, Yfl</i>	coordinates front left [m]
<i>Xfr, Yfr</i>	coordinates front right [m]
<i>Xrl, Yrl</i>	coordinates rear left [m]
<i>Xrr, Yrr</i>	coordinates rear right [m]
<i>Xtl, Ytl</i>	coordinates trailer left [m]
<i>Xtr, Ytr</i>	coordinates trailer right [m]

2. Dynamic parameters:

<i>SlopeLimit</i>	maximum rate at which the output values may change [m/s] (default: 1.1)
<i>EnableRefinement</i>	Default: 0, which means that bumps on the road are ignored. If this value is set to 1, the used road texture, as well as object strips and some objects (such as storm drains) will lead to bumps that can be seen in the Z and mu outputs.

3. Outputs:

<i>Zfl</i>	height front left [m]
<i>Zfr</i>	height front right [m]
<i>Zrl</i>	height rear left [m]
<i>Zrr</i>	height rear right [m]
<i>Ztl</i>	height trailer left [m]
<i>Ztr</i>	height trailer right [m]
<i>mufl</i>	friction coefficient front left [m]
<i>mufr</i>	friction coefficient front right [m]
<i>murl</i>	friction coefficient rear left [m]
<i>murr</i>	friction coefficient rear right [m]
<i>mutl</i>	friction coefficient trailer left [m]
<i>mutr</i>	friction coefficient trailer right [m]

4. Dependencies:

DPUVDSSCNX must run on the same computer as **DPUSCNX**. The *value* Index of **DPUVDSSCNX** must be higher than the value *Index* of **DPUSCNX**.



5. Functional principle:

DPUVDSSCNX collects the data of the ground height, and the friction coefficients, for up to six positions given through their X and Y coordinates of the global SILAB world coordinate system.

The outputs may be used for positioning of a graphical object, for example the ego vehicle. Additionally, they may be used to feed a driving dynamic simulation which relies on the height of individual wheels. (An example would be **DPUVDSST4WTR2W.**)

6. Multithreading mode:

In case **DPUVDSSCNX** runs in the same thread as **DPUSCNX**, complete access to the database is possible. That way for every position (X,Y) currently available in the database correct outputs can be made.

If **DPUVDSSCNX** is running in another thread the access to the database is limited. In this case the heights are only valid for a small area ($radius\ r = VehicleLength + HExtra$) around the position of the primary cursor (inputs X and Y of **DPUSCNX**). If inputs (X,Y) of **DPUVDSSCNX** point beyond this area the corresponding Z value will remain on its last value and a warning message is displayed in the SILAB log window.